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Uranium mining, health risks and environmental injustice

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Abstract

This article critically examines uranium mining as a global public and environmental health crisis, rather than merely a technical or industrial activity. It discusses how extractive operations produce cumulative, long-term harms ranging from radioactive exposure and ecological degradation to intergenerational health risks that fall most heavily on communities already marginalised by social and political inequalities. Drawing on the secondary sources of data and illustrative case studies, including those from Indigenous territories, it maps the deeply uneven distribution of risks and benefits across the uranium supply chain. These cases reveal how uranium extraction magnifies environmental injustice by (re)producing inequitable exposures, limiting community participation and denying recognition to affected peoples. We also interrogate the ethical pressure at the heart of contemporary sustainability debates: the promotion of nuclear energy as a climate solution while ignoring the toxic material realities on which it depends. As both a critical synthesis and a call for systemic reform, we argue for justice-oriented policies grounded in public health equity, environmental responsibility and genuinely participatory governance. Ultimately, rethinking uranium mining requires more than technological improvements; it demands confronting the structural injustices rooted in extractive economies and centring the lived experiences, rights and knowledge systems of the communities who bear the greatest burdens.

Keywords Uranium mining, Public health impacts, Environmental (in)justice, Radioactive contamination, Policy reform

1 Introduction

Nuclear power has long been promoted as a low-carbon alternative to fossil fuels, especially in the context of climate change mitigation and the global transition to sustainable energy [70]. Central to this energy pathway is uranium, a naturally occurring radioactive element, which plays a pivotal role in nuclear power generation. Globally, uranium is considered a key resource for achieving low-carbon energy goals. However, the extraction and processing of uranium ore are fraught with significant environmental, health and socio-political consequences. Industrial-scale uranium mining has led to severe environmental degradation, including air and water pollution, soil erosion, radioactive contamination, biodiversity loss and long-term challenges associated with tailings



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management [26]. It also disrupts ecosystems [8, 73, 103] and contributes to health burdens among the mine workers [3, 14] and particularly for communities located near mining sites [40, 56, 78, 90]. The mining and trade of uranium are shaped not only by geological availability and market dynamics but also by geopolitical agendas and international diplomatic priorities. As global demand for nuclear energy escalates, uranium mining has expanded rapidly, often resulting in severe environmental degradation and adverse health outcomes for local communities [59]. This expansion, however, is not merely a technological or energy-driven phenomenon; it reflects a deeper entrenchment of capitalist market logics that prioritize growth and resource extraction over ecological balance, public health and environmental justice. The nuclear energy sector, in this regard, perpetuates forms of development that are fundamentally at odds with environmental sustainability [62].

Exposure to radioactive materials and toxic byproducts has been linked to increased risks of cancer, respiratory illness and reproductive disorders. These health burdens often disproportionately affect marginalized and Indigenous populations, whose lands, livelihoods and bodies are systematically sacrificed in the name of national energy security and strategic interests [17, 44, 89, 90]. Brannon, [13] pointed out ‘bureaucratic violence,’ where the formal inclusion of justice language in environmental policy masks the continued marginalization and dispossession of uranium-affected populations. As such critiques endorse that ‘environmental injustice’ and ‘public health’ struggles in uranium-rich regions are not simply responses to isolated environmental harms, but rather symptomatic of systemic governance failures across regional, national and global scales. These failures manifest in weak enforcement, limited public participation and the erasure of Indigenous sovereignty in decision-making processes, thereby exacerbating the vulnerability of affected communities.

Uranium mining presents significant and long-term threats to both public and environmental health, yet much of the existing scholarship has remained focused on its technological, geological and industrial dimensions. Despite the global expansion of uranium extraction often legitimized through narratives of energy security and decarbonization the health-related and ecological consequences remain unevenly distributed and structurally under-addressed within mainstream policy frameworks. Many communities living near uranium sites come across prolonged exposure to radioactive contamination, air and water toxicity and inter-generational health burdens, which are often excluded from dominant discourses of risk management and corporate social responsibility. Contemporary uranium mining is not merely an economic enterprise or technical endeavour; it is implanted in an ongoing pattern of environmental injustice, disproportionately affecting Indigenous and marginalized communities. These spatialized harms raise urgent concerns around distributional, procedural and recognition injustices the core dimensions of the environmental justice framework [59]. Recent scholarship has thoroughly documented uranium’s status as an environmental-justice problem [61] and a focused assessment of the field has called for more integrated, comparative and political-economic analyses [59]. This article offers a justice-oriented comparative synthesis across three Indigenous contexts to identify recurring mechanisms of distributive, procedural and recognition failures that transcend national and regulatory boundaries.

The article analyses the dynamics through which Indigenous lands are transformed into ‘sacrifice zones,’ with particular attention to political and economic drivers, cultural

and spiritual relationships to land, the social and health consequences of displacement and three representative case studies: the Navajo Nation (United States), Jadugoda (India) and the Ranger mine on Mirarr Country (Australia). Through these examples, we illustrate how power imbalances have perpetuated environmental injustice and eroded Indigenous sovereignty, underscoring the urgent need for policy reform. In this article, we address uranium mining as a global public and environmental health crisis, rather than merely an environmental or industrial challenge. To substantiate this argument, we first offer a global overview of uranium extraction and demonstrates how extractive operations generate cumulative ecological degradation and chronic health risks across diverse socio-geographic settings. Drawing on case studies including those on Indigenous territories, it examines how regulatory failures, weak accountability mechanisms and environmental injustice intersect to reinforce socio-political marginalization and exclusion from decision-making. Our analysis also critically reflects on the ethical contradictions of nuclear energy in the context of sustainability and global climate agendas. Ultimately, we argue for urgent, equity-driven policy reforms grounded in public health principles, environmental responsibility, Indigenous and community rights and participatory governance. We adopt a justice-oriented analytical framework to examine uranium mining as a cumulative public and environmental health crisis. By referencing Indigenous contexts across the United States, India and Australia, we identify recurring distributive, procedural and recognition injustices embedded within extractive political economies.

1.1 Theoretical framework

We situate this article within contemporary environmental justice scholarship to analyse uranium mining as a structurally produced source of public-health risk and socio-political marginalisation. Environmental justice theory holds that environmental harms and benefits are not distributed randomly but are systematically patterned along axes of power, socio-economic status and marginalisation [82]. These patterns are produced and reproduced through institutional arrangements, political processes and socio-economic structures, directing attention not only to who bears harm but also to how governance, recognition and distribution shape those outcomes.

Scholars conceptualize environmental justice through multiple interrelated dimensions that function as analytical tools rather than purely descriptive categories. Distributive justice concerns the fair allocation of environmental burdens and benefits, highlighting how hazardous exposures and ecological degradation disproportionately affect marginalised populations, while economic and strategic gains flow toward more powerful actors. Procedural justice focuses on the fairness of decision-making processes and the meaningful involvement of affected communities in environmental governance, including access to information and participation in decisions that shape environmental outcomes [64, 82]. Recognition justice directs attention to the mis-representation and lack of representation of disadvantaged groups, emphasising the importance of acknowledging diverse voices, values and experiences in shaping policies that affect environmental conditions [82]. These three dimensions are analytically interdependent: durable distributive remedies are unlikely without procedural inclusion and cultural recognition and procedural reforms are weakened when they fail to attend to recognition claims [64].

Environmental justice cannot be fully understood without situating these dimensions within broader socio-political and economic structures that shape extractive resource governance. Research on extraction (especially uranium mining) and environmental justice shows how global commodity chains and institutional arrangements entrench inequities in exposure and access to resources, enabling economic value to be realised far from the frontiers where environmental harms occur [60]. For example, environmental sociology highlights how procedural and distributive inequities intersect in mining and resource extraction contexts, affecting marginalised communities' access to justice, participation and wellbeing [60]. In addition, environmental justice scholars argue that distributive justice cannot be attained or sustained without meaningful procedural engagement, such that genuine public participation is central to addressing unequal environmental burdens [64].

Integrating these dimensions within a structural lens allows us to connect local patterns of exposure and harm associated with uranium extraction especially in Indigenous territories to broader processes of institutional decision making and political-economic structure. This theoretical framework enables a justice-oriented analysis that identifies not only the uneven distribution of environmental harms but also the ways in which governance practices and recognition failures reproduce those inequities across space and time.

1.2 Research methods

We adopt a qualitative, theoretically guided and case-oriented research design to examine uranium mining through the lens of environmental justice in Indigenous contexts. We deliberately rely on a structured and systematic analysis of secondary sources. The long-term and cumulative harms associated with uranium extraction-radioactive exposure, ecological degradation and intergenerational health risks are documented across dispersed epidemiological studies, regulatory records, environmental assessments and community reports. A structured synthesis of these materials allows us to identify recurring patterns that may not be visible in isolated site-specific accounts. Our research is global in scope and argumentative in orientation. The case studies are not treated as self-contained empirical investigations; rather, they function as analytically illustrative sites that ground and substantiate a broader theoretical argument about environmental injustice in extractive economies.

We engage three Indigenous contexts affected by uranium mining: the Navajo Nation in the United States, Jadugoda in India and the Ranger mine region in Australia. These cases were selected purposively, not to construct a representative global sample, but to examine structurally comparable settings where uranium extraction has occurred in or adjacent to Indigenous territories and where documented evidence exists regarding environmental contamination, health risks and governance contestation. Each site reflects the incorporation of Indigenous lands into national nuclear programmes under asymmetrical power relations. Despite differences in regulatory systems and political contexts, the cases share key features: documented radioactive exposure, long-term ecological and health concerns and sustained claims of limited participation or marginalisation in decision-making processes. Their analytical value lies in demonstrating how similar patterns of cumulative harm and environmental injustice emerge across distinct national settings. These cases therefore function as illustrative reference points

within a broader global argument. They present the theoretical analysis in empirically documented contexts while enabling examination of recurring structural dynamics that extend beyond any single site.

We employed secondary sources and advance a justice-oriented analytical framework. The documentary corpus was assembled through a structured search of Google Scholar, Scopus and relevant institutional repositories for materials published between 1980 and 2025. The search employed combinations of keywords including “uranium mining,” “Indigenous communities,” “environmental justice,” “radiation exposure,” “health impacts,” “contamination,” as well as the specific case locations (Navajo Nation, Jadugoda and Ranger mine region). In total, 89 documents were identified, screened and included in the analysis. The corpus comprises peer-reviewed journal articles, epidemiological studies, environmental monitoring reports, governmental and regulatory documents and reports produced by Indigenous organisations and civil society groups. Source selection was guided by analytical relevance to uranium extraction, documented health risks, environmental contamination and governance processes.

Data analysis is guided by a justice-oriented framework that interprets health risks and environmental impacts not as isolated technical outcomes, but as socially structured and politically mediated processes. We organize the analysis around three inter-related dimensions of environmental justice (distributive, procedural and recognition justice) which serve as the central organising principles of interpretation. Through this structured interpretive synthesis, we connect empirical documentation to a broader theoretical argument concerning the patterned production of environmental injustice in uranium mining regions.

1.3 Uranium mining industry: a global overview

The global uranium industry has evolved through a complex interplay of geopolitical, technological and energy-sector transformations. What began as a strategic resource race during the early nuclear age has gradually expanded into a broader and often contested framework of energy security, climate mitigation and global supply-chain politics. Understanding this trajectory is essential for analysing contemporary debates surrounding nuclear power, resource governance and the socio-environmental consequences of extraction. Uranium extraction began in the mid-twentieth century, largely driven by military imperatives especially the nuclear weapons programmes of the United States and the United Kingdom, with South Africa serving as a major supplier during this period [65]. Owing to its exceptionally high energy density, uranium soon became central to the expansion of civilian nuclear power, playing a crucial role in the energy strategies of many countries [80]. Framing uranium solely as a commodity disguises its position within contested national and transnational debates about energy and security. Global critiques of nuclear power [91] and historical studies of nuclear harms (Alexis-Martin, 2019) underline that extraction must be situated within wider political and economic debates about nuclearity and energy transitions.

Over the past seven decades, uranium has been used both for energy production and for various military applications. The enrichment process, carried out to produce fuel-grade uranium, simultaneously generated depleted uranium (DU) as a by-product, which has subsequently been utilised in radiation shielding, armour-penetrating munitions and as ballast in aircraft [30]. As global energy demand increased and concerns

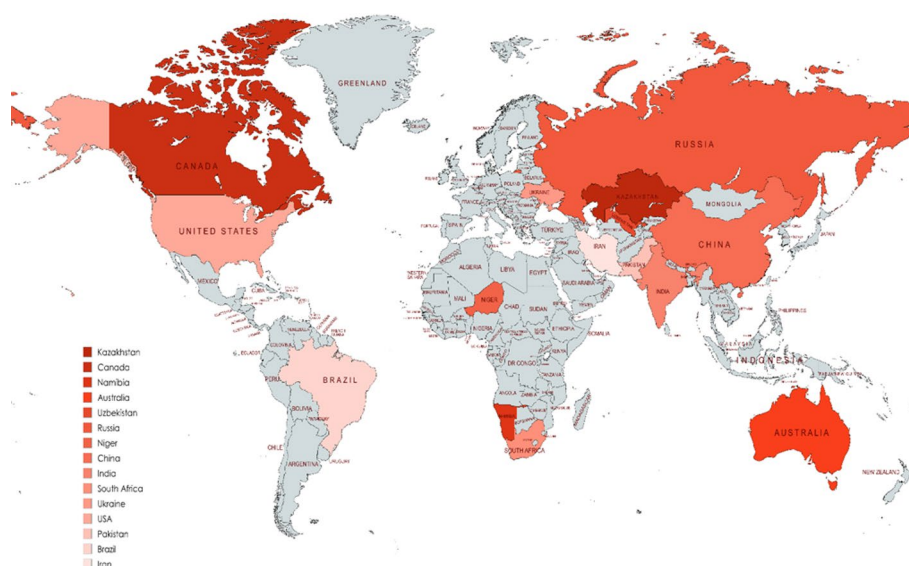


Fig. 1 World Uranium production (2022). Source (data): world nuclear association [104]

about climate change intensified, uranium mining gained renewed significance, with many states reframing nuclear power as a low-carbon alternative [25]. Australia possesses the world's largest identified uranium reserves, a result of its favourable geological formations [45]. Kazakhstan, meanwhile, has emerged as the world's leading producer and exporter, shaping global nuclear fuel supply chains and consolidating its strategic position in the industry [97]. Despite these developments, the global uranium market continues to confront a widening supply-demand imbalance, underscoring the need for long-term planning and sustainable resource governance [84]. Africa's uranium sector also attracts substantial international interest, in part because of comparatively relaxed regulatory frameworks and lower labour costs, which together create favourable conditions for investment though often at the expense of environmental and social protections [25]. Recent production patterns further underscore this concentration of supply. In 2022, global uranium output reached approximately 49,355 metric tons meeting nearly three-quarters of total global demand. Production continues to be dominated by a handful of countries, with Kazakhstan, Canada and Australia collectively accounting for almost two-thirds of worldwide output (Fig. 1). Australia holds the largest share of identified uranium reserves (around 28% of the global total), while Kazakhstan, despite possessing about 13% of known reserves, has emerged as the leading producer, supplying roughly 43% of global uranium in 2022 (World Nuclear Association, [104]).

The discourse and data on global uranium mining sector reveal not only a highly concentrated and strategically sensitive resource economy but also one marked by deep socio-environmental contradictions. While uranium is increasingly positioned as a low-carbon fuel capable of supporting long-term energy security, its extraction continues to impose disproportionate public health risks, ecological degradation and radiological exposure on mining-dependent regions. As demand for uranium rises amid global decarbonisation efforts, these unresolved tensions highlight the urgent need for more equitable governance frameworks that prioritise community rights, long-term ecological sustainability and robust public health protections.

1.4 Environmental degradation and health effects

In the early decades of uranium mining, improper disposal of mill tailings (without liners) led to substantial contamination of soil, surface water and groundwater, without effective tailings control, disposal sites were widespread long-term failures of containment structures include the dispersion of radioactive dust from dry tailings, erosion of the containment structure and the release of waste materials from the outer surface of the containment structure and are a major cause of environmental impacts [1, 71]. However, uranium and aluminum elements were accelerated to migrate into the soil aqueous phase and become bioavailable, in-situ measurements showed that the feeding activity of edaphic organisms assessed by the bait lamina assay was reduced, especially in soils adjacent to open mines, where leaching of stockpiled materials was carried out and where sludge from waste treatment plants was discharged [5, 71]. Furthermore, damage to the soil microbial community was also recorded through the assessment of soil enzyme activity and the analysis of other biomarkers of metabolically active components of the soil microbial community, such as phospholipid fatty acids and ergosterol, extracted from soil samples collected at the same location [6, 38, 71]. Considering the role of edaphic fauna and soil microbial communities in the mineralization of organic debris, which is one of the most important functions of soil, these results unequivocally confirm that this fundamental function of the soil in this region was probably seriously impaired by contamination with acids, metals and the role of edaphic fauna and soil microbial communities in the mineralization of organic debris, which is one of the most important functions of soil, these results unequivocally confirm that this fundamental function of the soil in this region was probably seriously impaired by contamination with acids, metals and radionuclides [71]. Waste dumps and tailings ponds associated with mining residues contain deposits of material derived from overburden, valuable minerals below cut-off grade, or material rejected in the concentration process and worldwide, thousands of abandoned mines have left behind tailings and tailings ponds and these increase the flow of water from the mines into waterways, which contains heavy metals and other pollutants [21].

Mining results in three main types of changes: changes in the natural topography which limit the potential for other uses of the land, changes in hydrogeological conditions which affect both groundwater and also change in the geotechnical conditions of the rock, these changes caused by mining can have a variety of impacts on the geo-environment, impact on the Hydrosphere (include the depletion of groundwater levels, the discharge of mine water into rivers, seas and lakes, leakage from standing tanks and evaporators that negatively affect groundwater quality and the pumping of water into the ground to extract minerals, depending on the chemical composition of the rock, the extracted water is usually highly acidic, causing the dissolution of a variety of toxic and heavy metal compounds), impact on the atmosphere (mining occur not only from the internal combustion engines of mining equipment, but also from explosions, rock and mineral deposits and dust and gas emissions), impact on the biosphere (mainly through pollution and loss of biodiversity through land, vegetation degradation and local micro-climates) [67].

Uranium ores often contain sulphide minerals and when exposed to air and water during mining, these sulphides oxidize, forming sulfuric acid and this acidic flow dissolves heavy metals such as arsenic, cadmium, lead, nickel, zinc and radionuclides from the

surrounding rock, creating highly toxic acid rock drainage or acid mine drainage (Lottermoser, [55]). Waste rock piles, low-grade ore and overburden and finely ground ore residues left after uranium extraction are major, long-term sources of pollution. Rainwater flowing through these materials leaches' radionuclides, particularly radium-226, uranium and thorium-230, heavy metals and residual processing chemicals such as sulfuric acid, ammonia, solvents and other contaminants into groundwater [1]. Modern regulations require treatment, but spills, leaks and inadequate treatment remain risks [15]. Mining operations generate large amounts of dust from excavation, blasting, excavation, transportation and expose waste rock/tailings and this dust contains radioactive particles, uranium, thorium, radium decay products and potentially toxic heavy metals such as arsenic, lead, air dispersion can contaminate soil, vegetation and surface water over considerable distances ([1, 21];). Compaction by heavy machinery, damage to soil structure, removal of organic matter and acidification severely degrade soil quality, hindering the recovery of natural vegetation and ecosystems [85]. Impacts include: acute toxicity to fish and invertebrates, bioaccumulation and biomagnification of pollutants in the food chain, disruption of reproductive cycles, loss of sensitive species and overall reduction in biodiversity and ecosystem functioning. Soil contamination and physical disturbance directly affect flora and fauna, plants can take up radionuclides and heavy metals, causing toxicity, growth retardation and contamination of herbivores [86]. Pollution can significantly alter microbial communities in soils and sediments, which are crucial for nutrient cycling and ecosystem functioning, but the immobilization of some microbial contaminants can facilitate biological remediation, high levels of toxicity can suppress microbial activity [1]. Long-term institutional control and monitoring are essential but create financial and administrative burdens for future generations [20].

Ethnographic and epidemiological studies have repeatedly documented long-term contamination and associated health burdens in uranium-affected communities. Oral-history and community studies on the Navajo Nation record persistent soil and water contamination alongside chronic health complaints [19]. Detailed analyses of mill tailings exposures further demonstrate enduring pathways of risk long after operations cease [58], a pattern mirrored in recent regional reviews of reservation-adjacent legacy sites [66].

Mining is one of the most dangerous industrial occupations and during the period 1980-89, the mining industry had the highest annual number of injury deaths [67]. The production of large quantities of waste materials has become a cause of environmental and health concerns [21]. The health effects of mining can be divided into two categories: immediate effects such as accidents; and cumulative and progressive effects such as stress, radiation and lung disease (decades of research on the incidence and risk of lung cancer in miners. In general, the risk of radon emitted from tap water and inhaled is greater than that of radon ingested through drinking water) [7, 67]. Animal studies and human data from miners and workers accidentally exposed to uranium have shown that uranium primarily affects the proximal tubules of the kidneys. At very high doses, tubular degeneration and necrosis (i.e., tissue death) can occur within days of uranium intake [83].

The toxicity and radioactivity of uranium are major environmental concerns in the exploitation and processing of U-ores. In areas where U mining is planned, it is recommended to conduct a background assessment of harmful elements and a careful

assessment of the concentrations of toxic and radioactive elements before starting mining operations to prevent future contamination [22]. Health effects from mining can be divided into two categories: immediate effects such as accidents; and cumulative and progressive effects such as stress, radiation and lung disease, in the term of health risks, four different types: Physical hazards (include noise, heat, vibration, falls and explosions, floods and various types of dust, aerosols and fine particles which cause fibrogenic and carcinogenic effects), Chemical Hazards (arise from chemical pollutants in water, solid waste and air, the most common of which are carbon monoxide and dioxide, oxides of sulphur, nitrogen oxides and fluorine compounds), Biological hazards (where hygiene and sanitation standards are poor, miners are more likely to be exposed to biological hazards caused by living organisms such as fungi, bacteria and parasites), psychological risks (risks associated with mining include claustrophobia, anxiety, tension or irritability associated with awareness of hazardous work environments) [67] and radiological hazards (universally identifies as the most significant radioactive hazard in underground uranium mining) [53]. It can be detected by ingestion and inhalation of ^{222}Rn in drinking water and indoor air, Knowledge of soil gas and underground radon levels therefore, drinking water is important to protect the public from the consequences of excessive exposure to radon, assessing drinking water quality through radon assessment key aspect research of public health [2].

Although alpha radiation from these isotopes is less dangerous externally, internal deposition in the lungs and transfer to other organs such as the bones and liver leads to long-term internal radiation and potential chemical toxicity [15]. Gamma radiation also from many years chains also contribute to external exposure, gamma radiation from the ores and waste rock, Silicosis, a progressive and incurable fibrotic lung disease, remains a major occupational health risk, silicosis also increases susceptibility to tuberculosis and lung cancer [54, 77, 94]). High exposure can cause severe kidney damage, including tubular necrosis and possible acute kidney failure [51]. Chronic low-level exposure, although less clearly associate with overt renal failure in human occupational studies than acute toxicity in animal models, is still a significant concern, potentially contribute to an increase risk of subclinical kidney dysfunction and other renal influents [49]. Exposure to these substances can have their own health effects, including neurological damage, heart disease and additional organ toxicity [18]. Exposure to dust, both silica, common mining dust and diesel emissions, contributes to the development of chronic bronchitis and emphysema [37]. Epidemiological studies in occupationally expose groups have struggle to demonstrate a clear excess of chronic kidney disease or end-stage renal disease that is directly attributable to exposure to uranium at typical mining levels alone, partly due to some factors such as hypertension, diabetes and long latency [49]. Potential effects on the reproductive system, cardiovascular system and immune system have been investigated, but evidence is limited and often inconclusive in human occupational studies, complicate by multiple exposures and lifestyle factors [15]. These communities often bear disproportionate burden due to proximity, cultural practices, livelihoods, historical neglect and environmental injustices, they face heightened risks through multiple pathways and often lack sufficient resources or political power to demand mitigation [14].

1.5 Uranium mining in indigenous territories: insights from case study

Indigenous communities have lived on and protected their ancestral lands for thousands of years. However, these lands are now increasingly being exploited by large-scale mining industries that use advanced technologies for extraction, transportation and marketing. The impact of such mining is often massive reshaping entire landscapes by removing mountains and diverting rivers. For Indigenous peoples, these places are not just physical spaces but hold deep cultural and spiritual meaning, believed to be shaped and inhabited by ancestral beings. Indigenous territories contain a significant share of global mineral resources. As a result, mining activities in these areas frequently lead to the erosion of Indigenous cultures, displacement of communities and degradation of both the environment and traditional ways of life [68]. These Indigenous territories have long been treated as ‘sacrifice zones’¹ in the global pursuit of uranium, a trend driven by their systemic social and political exclusion [95]. This marginalization facilitates environmental destruction and public health crises. However, as comparative and multidisciplinary research reveals, the impacts of mining are not uniform. Indigenous communities actively engage with these extractive projects resisting, negotiating, or adapting depending on political and legal contexts [47, 56, 57, 74].

In 1948, the U.S. Atomic Energy Commission initiated uranium extraction on Navajo lands, first near Cove, Arizona, then expansive across the reservation. By 1956, more than 750 mines operated under minimal safety protocols and by cessation in 1967, over 1,000 abandoned mine sites scarred the landscape. Tailings left in open pits leached into aquifers; livestock drinking contaminated water became vectors of radioactive dust into homes. Research has since implicated uranium-related metals in cardiopulmonary toxicity, immune dysfunction and oxidative stress among residents exposed to mine wastes [10, 41]. Despite decades of activism and partial remediation efforts, many Navajo families still lack access to clean water and trust in federal agencies remains fractured. The Navajo case presents classic evidence of legacy contamination and contested remediation; community testimony and synthesis volumes document the social and intergenerational consequences of mine abandonment [16]. Ethically attentive research practices are required given the history of problematic research in this terrain [69].

In India's Singhbhum belt, Jadugoda's uranium mines, operated since 1968 by the Uranium Corporation of India Limited, echo this pattern of dispossession and health impacts. Nestled within dense forests and riverine ecosystems, the seven major Jadugoda deposits supplied the nascent Indian nuclear program, yet their tailings ponds where slurry laden with radionuclides and heavy metals sits in open air have overflowed into paddy fields and wells relied upon by *Adivasi* (Tribal) communities [34, 39]. Despite repeated appeals, compensation schemes remain narrowly defined, covering only direct medical costs and ignoring losses of traditional ecological knowledge and cultural practices tied to ancestral forests. As Majumdar et al. [56] reported widespread chronic health issues in villages near the uranium mining site, including respiratory and skin diseases, cancers, neuropsychiatric disorders, birth defects and reproductive problems. Psychological distress such as anxiety, fear and suicidal thoughts is also common. Locals identify unprotected tailing ponds as a key source of radiation exposure. The Jadugoda

¹ Certain regions and communities are treated as disposable in the name of development, where their displacement, environmental degradation, health hazards, and loss of livelihood are rendered acceptable or overlooked for the sake of resource extraction and economic gain. It perpetuates not only dispossession but also emerges as a symbol of resistance and collective struggle (See, [95]).

experience therefore reveals how developmental priorities framed in national security rhetoric can override not merely property rights, but the very lifeways of Indigenous peoples, entrenching social inequities that far outlast mining operations. Though embedded in a different regulatory and historical setting, Jadugoda exhibits similarly persistent exposure pathways and contested governance dynamics; comparing Jadugoda to Navajo highlights how distributive and procedural failures recur under different institutional forms.

In Australia's northern territory, the Ranger uranium mine has been a focal point of controversy and Indigenous opposition since its proposal in the 1970s. Situated on the ancestral lands of the Mirarr Traditional Owners, approximately 250 km east of Darwin, the mine has been operational since 1980 under the management of Energy Resources of Australia. According to Hein [42] and Waggitt [101], tailings are chemically neutralised before being deposited in a designated storage facility. However, mining operations typically involve extensive land clearance to facilitate access to ore deposits and to construct associated infrastructure such as access roads, tailings dams, milling facilities, waste rock dumps and open-cut pits. This land clearance represents one of the most visible and ecologically damaging impacts of mining. In countries like Madagascar, such clearance is recognised as a leading cause of forest habitat loss, degradation and fragmentation [46, 100, 102]. The Ranger region demonstrates another permutation of extractive governance in a settler-colonial context, where regulatory frameworks and Indigenous rights claims produce distinct but related forms of marginalisation.

The transformation of Indigenous territories into uranium sacrifice zones is neither incidental nor inevitable; rather, it reflects a historically entrenched logic of colonial dispossession and systemic devaluation of Indigenous life [32]. These zones demonstrate the extractive tension wherein value is created for dominant systems through the simultaneous production of waste and harm, disproportionately offloaded onto marginalized communities. As Dunlap and Jakobsen [29] argue, capitalist modernity operates through evolving technologies and discursive mechanisms that enable states and corporations to expand the frontiers of extraction, often by bypassing or weakening regulatory and cultural safeguards. This dynamic is clearly reflected across the three case studies: governments and corporations identify uranium-rich Indigenous lands, strategically manipulate legal and policy frameworks, extract resources with minimal concern for long-term safety and ultimately leave behind toxic legacies ranging from radioactive tailings to degraded ecosystems and health burdens that persist across generations. These sacrifice zones not only disrupt ecological and social reproduction but also erase Indigenous ontologies and epistemologies, undermining deep relationships with land and knowledge systems [32]. Yet, as Castán Broto and Sanzana Calvet [23] emphasize, these are not merely spaces of abandonment they are also sites of resistance, where affected communities demand justice, recognition and structural change. Responding to this enduring cycle of extraction and abandonment requires transformative policy action: the implementation of legally binding Free, Prior and Informed Consent (FPIC), the establishment of comprehensive remediation bonds that account for the full lifecycle of uranium extraction, mandatory health and environmental assessments led by affected communities and statutory recognition of intangible cultural heritage.

1.6 Environmental injustice and political marginalization

The dynamics of uranium extraction illustrate how political marginalisation, social injustice and environmental injustice intersect in resource-rich regions. These territories often remain politically voiceless while being embedded in expansive trade networks, foreign investment flows and strategic diplomatic arrangements [99]. Even as policies attempt to legitimise uranium mining, extractive regimes continue to reproduce structural negligence and hierarchical domination [27, 63]. Uranium-rich communities face political marginalisation through economic dependency, political subordination, cultural displacement, health risks and systemic invisibility [24, 35, 43]. For Indigenous groups, these conditions translate into multiple, overlapping injustices: (i) distributive injustice: state gains versus community harms; (ii) procedural injustice exclusion and absence of free, prior and informed consent [50]; (iii) recognition injustice denial of cultural and territorial rights [52]; (iv) environmental injustice extraction practices that exceed or violate agreed limits [43, 98]. The continuation of these injustices is closely tied to how extraction is institutionalised through geopolitical, legal and regulatory arrangements. Uranium extraction has evolved into a geopolitical strategy embedded within trade agreements, treaties and regulatory frameworks, placing it under direct political jurisdiction [36]. Yet gaps in regulation and corporate impunity allow extraction to proceed without accountability, producing conditions of political marginalisation, civil injustice and environmental injustice [61]. Weak enforcement of impact assessments, overreliance on general legal frameworks and local political alliances further sideline community protection and diminish Indigenous participation in decision-making processes [33]. These governance structures institutionalise political marginalisation by bypassing community consent, suppressing Indigenous claims and enabling state power to legitimise extraction without recognition of Indigenous authority [4]. Compensation packages, resettlement promises and livelihood assurances often function as tools of pacification, masking deeper exploitation [48, 96]. This leads to civil and distributive injustice, reflected in excessive extraction, minimal labour protection for underpaid Indigenous workers and negligence toward threat management, health risks and socio-demographic disruptions [56, 92]. Simultaneously, environmental injustice is reproduced through the absence of robust environmental assessments, political shielding of extractive companies, bribery and negotiated regulatory leniency, enabling continued ecological degradation [81]. In some cases, extreme poverty pushes Indigenous communities to relinquish land in exchange for short-term compensation, further deepening their vulnerability.

In continuation of these entrenched forms of political marginalisation and environmental injustice, Indigenous communities and allied organisations have increasingly mobilised to contest the uranium extraction regime. The severe hardship, systematic oppression and impoverishment experienced at extraction sites have long drawn attention from NGOs, (Non-Government Organizations) scholars, health practitioners and critical researchers [24]. This visibility has facilitated the emergence of dense activist networks community-based organisations, Indigenous movements and NGO-community collaborations dedicated to knowledge sharing, advocacy and protection [12, 93]. While early activism primarily targeted corporate misconduct and its damaging environmental and health effects, expanded support from national and international NGOs enabled a sharper critique of the state's complicity—evident in policy exemptions, regulatory inadequacies and the systematic suppression of dissent (Conde, 2015). As a result,

activists increasingly frame uranium mining as a human rights violation and an environmental justice crisis, rather than an economic or technical enterprise [11]. Yet, despite persistent advocacy, governments and corporations routinely ignore the suffering and claims of affected communities [76]. Their responses minimal penalties, enhanced compensation packages, or temporary suspension of operations function largely as mechanisms of containment, offering superficial relief while safeguarding extractive interests [75]. These practices not only perpetuate structural injustice but also reinforce the political marginalisation of Indigenous peoples within extraction zones.

Earlier scholarships (for example, [57, 61]) demonstrate that affected communities sometimes endorse mining amid harm, while this pattern is not merely a distinctive phenomenon but surfaces across divergent governance regimes, albeit mediated by distinct local histories, modes of recognition and institutional architectures. Malin & Alexis-Martin [59] call for broader, integrative research agendas, our findings operationalize that call by applying distributive, procedural and recognition lenses across three Indigenous contexts, thereby showing recurring mechanisms cumulative exposure, procedural exclusion and epistemic marginalisation that reproduce injustice across political boundaries. Contextualizing, Fegadel's [31] critique of toxic colonialism: where Fegadel frames uranium impacts in explicitly colonial and genocidal terms, our justice-oriented analysis situates such claims within a comparative framework that highlights both shared colonial dynamics and variant institutional forms of redress and resistance.

Our analysis of uranium extraction in Indigenous territories shows that political marginalisation, social injustice and environmental injustice are rooted within the broader architecture of extractive governance. Uranium mining operates through alliances between state institutions, corporate actors and geopolitical interests, producing conditions in which Indigenous rights, health and environments are regularly compromised. Regulatory gaps, selective enforcement and policy exemptions allow extraction to proceed with limited accountability, while the heaviest burdens cultural disruption, health risks, labour precarity and ecological degradation fall disproportionately on Indigenous communities. These patterns expose injustice not as an accidental by-product but as a structural consequence of extraction. At the same time, Indigenous mobilisation provides a critical challenge to these entrenched inequities. Community organisations, activist networks and transnational NGOs have reframed uranium mining as a human rights and environmental justice issue, highlighting the inadequacies of state protection. Yet government and corporate responses often remain superficial, offering compensation or temporary suspensions that leave underlying power imbalances intact. Uranium extraction is fundamentally political, reshaping territorial authority, decision-making power and environmental futures. Addressing its impacts requires structural reforms grounded in Indigenous sovereignty, sustainability principles and robust public health accountability.

2 Conclusion

By engaging deeply with the uranium mining and environmental justice related literatures [16, 31, 59, 61] and by synthesising cross-case documentary evidence under an explicit justice framework, we advance the field by demonstrating that justice deficits in uranium extraction are structurally patterned and demand policy remedies that centre recognition, procedural equity and long-term health monitoring. The ethical

debates surrounding nuclear energy are intensified by the disproportionate distribution of its benefits and harms. While nuclear power contributes significantly to national and regional electricity generation, the sacrifices demanded of affected communities through displacement, post-operational neglect, demographic shifts and enduring social-environmental burdens reveal deep inequities embedded in energy policy [72, 88]. The persistent absence of procedural justice, informed consent and intergenerational responsibility reflects an extractivist logic that treats communities and ecosystems as possessing unlimited assimilative capacity [28]. Such inequitable utilisation of resources positions uranium extraction as a transition marked by exploitation rather than genuine socio-environmental progress [79]. Emerging global norms decarbonisation, participatory decision making and non-coercive processes in land acquisition should therefore reshape extractive policy frameworks [9]. Yet the continued deployment of uranium mining as an instrument of national politics and international relations undermines redistributive justice [87]. Strengthening international treaties and protective mechanisms may offer more effective safeguards than localised resistance alone [36]. Ethically rethinking nuclear energy requires: (i) full disclosure of environmental risks, ecological impacts and mitigation measures rather than obscuring harms; (ii) meaningful reparations adequate compensation, resettlement and recognition of demographic disruption; and (iii) strong, independent oversight bodies insulated from political and corporate influence. These steps are central to aligning nuclear policy with principles of justice, sustainability and ethical governance.

Ultimately, rethinking nuclear energy requires confronting the ethical contradictions at the heart of contemporary energy transitions. Any vision of a sustainable future is fundamentally incomplete if it (re)produces the same hierarchies of power, marginalisation and ecological sacrifice that have long shaped extractive industries. Without institutional reforms grounded in justice, transparency and Indigenous rights, nuclear energy's claim to being "clean" remains ethically untenable. The challenge, therefore, is not only technological but deeply political: to design energy systems that distribute risks and benefits equitably, honour intergenerational responsibility and safeguard vulnerable communities. Only through such a transformation can nuclear energy be situated within a genuinely just and sustainable development trajectory.

Acknowledgements

The authors would like to thank the two anonymous reviewers and editorial team members for their constructive comments and valuable suggestions, which helped improve the quality of this manuscript.

Author contributions

KM: Conceptualization, design, data collection, analysis and manuscript drafting; LC and SP: Data collection, analysis and manuscript drafting; SHP: Data collection. All authors reviewed and approved the final manuscript.

Funding

Authors did not receive any funding for this study.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethical approval and consent to participate

This study is based exclusively on secondary sources, including peer-reviewed publications, publicly available reports and archival documents. It does not involve human participants, personal data or primary data collection. Therefore, formal ethical approval was not required. The research has been conducted in accordance with established academic standards of integrity and ethical scholarship.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 18 November 2025 / Accepted: 28 April 2026

Published online: 17 May 2026

References

1. Abdelouas A. Uranium mill tailings: geochemistry, mineralogy, and environmental impact. *Elements*. 2006;2(6):335–41.
2. Ajiboye Y, Isinkaye MO, Badmus GO, Faloye OT, Atoiki V. Pilot groundwater radon mapping and the assessment of health risk from heavy metals in drinking water of southwest, Nigeria. *Heliyon*. 2022;8(2):e08840. <https://doi.org/10.1016/j.heliyon.2022.e08840>.
3. Albrecht H. The heritage of uranium mining in the German-Czech ore mountains. *Entrepreneurship History*. 2017;2:88–106.
4. Altman J. Indigenous rights, mining corporations, and the Australian state. *The Politics of Resource Extraction: Indigenous Peoples, Multinational Corporations and the State*. London: Palgrave Macmillan UK; 2012. pp. 46–74.
5. André A, Antunes SC, Gonçalves F, Pereira R. Bait-lamina assay as a tool to assess the effects of metal contamination in the feeding activity of soil invertebrates within a uranium mine area. *Environ Pollut*. 2009;157(8–9):2368–77.
6. Antunes SC, Pereira R, Marques SM, Castro BB, Gonçalves F. Impaired microbial activity caused by metal pollution: a field study in a deactivated uranium mining area. *Sci Total Environ*. 2011;410:87–95.
7. Appleton JD. Radon: Sources, Health Risks, and Hazard Mapping. *AMBIO: A Journal of the Human Environment*, 2007;36:85–89. <https://doi.org/10.1579/0044-7447>.
8. Balaram V, Rani A, Rathore DPS. Uranium in groundwater in parts of India and world: A comprehensive review of sources, impact to the environment and human health, analytical techniques, and mitigation technologies. *Geosyst Geoenvironment*. 2022;1(2):759–78. <https://doi.org/10.1016/j.jafrearsci.2016.12.004>.
9. Béland D, Campbell B, Coderre M, Haang'andu P. Policy change and paradigm shifts in Sub-Saharan Africa: implementing the Africa Mining Vision. *Can J Afr Studies/Revue canadienne des études africaines*. 2022;56(1):79–97.
10. Blake JM, Avasarala S, Artyushkova K, Ali AMS, Brearley AJ, Shuey C, Cerrato JM. Elevated concentrations of U and co-occurring metals in abandoned mine wastes in a northeastern Arizona Native American community. Volume 49. *Environmental science & technology*; 2015. pp. 8506–14. 14.
11. Borde R. Protecting indigenous land from mining: a study of activist representations of indigenous people, in the context of anti-mining movements, with a focus on an Indian case (Doctoral dissertation, Wageningen University and Research). 2017.
12. Branagan M. The Australian movement against uranium mining: Its rationale and evolution. *Int J Reg Rural Remote Law Policy*. 2014;1:23–34.
13. Brannon MM. Uranium mining remediation, environmental assessment, and bureaucratic violence. *Environ Sociol*. 2024;11(2):186–97. <https://doi.org/10.1080/23251042.2024.2417443>.
14. Brugge D, Goble R. The history of uranium mining and the Navajo people. *Am J Public Health*. 2002;92(9):1410–9.
15. Brugge D, deLemos JL, Oldmixon B. Exposure pathways and health effects associated with chemical and radiological toxicity of natural uranium: a review. *Rev Environ Health*. 2005;20(3):177–94.
16. Brugge D, Benally T, and Esther Yazzie-Lewis. *The Navajo People and Uranium Mining*. University of New Mexico; 2006.
17. Brugge D, Benally T, Yazzie-Lewis E, editors. *The Navajo people and uranium mining*. UNM; 2007.
18. Brugge D, Buchner V. Health effects of uranium: new research findings. 2011.
19. Brugge D, Missaghian M. Protecting the Navajo people through tribal regulation of research. *Science and engineering ethics*. 2006;12(3):491–507. <https://doi.org/10.1007/s11948-006-0047-2>.
20. Bugai DO, Laptev GV, Skalsky OS, Lavrova TV, Avila R. Analysis of spatial distribution and inventory of radioactivity within the uranium mill tailings impoundment. *Nuclear Physics and Atomic Energy*. 2015;16(3):254–62. <https://doi.org/10.15407/jnpae2015.03.254>.
21. Candeias C, Ávila P, Coelho P, Teixeira JP. Mining activities: health impacts. *Reference module in earth systems and environmental sciences*. 2018; 1–21.
22. Carvalho FP. Environmental remediation and the legacy of uranium mining waste in Portugal and Europe. *Lessons to retain*. *Advanced Materials Research*. 2010;107:157–61.
23. Castán Broto V, Sanzana Calvet M. Sacrifice zones and the construction of urban energy landscapes in Concepción, Chile. *J Political Ecol*. 2020;27(1):279–99. <https://doi.org/10.2458/v27i1.23059>.
24. Conde M, Kallis G. The global uranium rush and its Africa frontier. Effects, reactions and social movements in Namibia. *Glob Environ Change*. 2012;22(3):596–610. <https://doi.org/10.1016/j.gloenvcha.2012.03.007>.
25. Dasnois N. Uranium mining in Africa: a continent at the centre of a global nuclear renaissance. 2012.
26. Davies WTR. Making sense of complex socio-ecological issues: a frame-analysis of Arctic natural resource development (Doctoral dissertation, University of Leeds). 2018.
27. DeBoom MJ. Nuclear (geo) political ecologies: A hybrid geography of Chinese investment in Namibia's uranium sector. *J Curr Chin Affairs*. 2017;46(3):53–83.
28. De Bruyn B. Introduction: beyond the sacrifice zone. *Textual Pract*. 2023;37(10):1475–98.
29. Dunlap A, Jakobsen J. The violent technologies of extraction, Palgrave Macmillan. 2020; pp. 91–118 <https://doi.org/10.1007/978-3-030-26852-7>.
30. Ebaid YY. Use of gamma-ray spectrometry for uranium isotopic analysis in environmental samples. *Rom J phys*. 2010;55(1–2):69–74.
31. Fegadel AR. Green victimization of native Americans: Uranium mining as a form of toxic colonialism and genocide. *Crit Crim*. 2023;31(2):489–505. <https://doi.org/10.1007/s10612-022-09679-0>.
32. Gómez-Barris M. *The extractive zone: Social ecologies and decolonial perspectives*. Duke University Press; 2017. <https://doi.org/10.1215/9780822372561>.

33. Göcke K. Indigenous peoples in the nuclear age: Uranium mining on indigenous' lands. In: *Nuclear Non-Proliferation in International Law - Volume I*. The Hague: T.M.C. Asser Press. 2014;199–223. https://doi.org/10.1007/978-94-6265-020-6_8.
34. Goswami JK. Struggle for the identity of tribal: A case study through the light of documentary Buddha Weeps in Jadugoda by Shriprakash. 2019.
35. Graetz G. Uranium mining and First Peoples: the nuclear renaissance confronts historical legacies. *J Clean Prod*. 2014;84:339–47.
36. Graetz G, Manning H. The politics of uranium mining in Australia. *Australia's Uranium Trade*. Routledge; 2016. pp. 137–64.
37. Grahame TJ, Schlesinger RB. Health effects of airborne particulate matter: do we know enough to consider regulating specific particle types or sources? *Inhalation Toxicol*. 2007;19(6–7):457–81.
38. Guedes MJ, Pereira R, Duarte K, Rocha-Santos TA, Antunes SC, Goncalves F, Freitas AC. Sterols and fatty acid biomarkers as indicators of changes in soil microbial communities in a uranium mine area. *J Environ Sci Health Part A*. 2011;46(6):659–68.
39. Gupta R, Sarangi AK. Emerging trend of Uranium mining: the Indian scenario. In *IAEA symposium on Uranium production and raw materials for the nuclear fuel cycle—Supply and demand, economics, the environment and energy security*, Vienna. 2005; (pp. 47–56).
40. Haakonde T, Yabe J, Choongo K, Chongwe G, Nchima G. Uranium exposure and health risk implications: A preliminary study among the residents living around uranium mining sites in the Southern Province of Zambia. *Environ Adv*. 2021;5. <https://doi.org/10.1016/j.envadv.2021.100098>.
41. Hoover JH, Bolt AM, Burchiel SW, Cerrato JM, Dashner-Titus EJ, Erdei E, Lewis JL. A Transdisciplinary Approach for Studying Uranium Mobility, Exposure, and Human Health Impacts on Tribal Lands in the Southwest United States. *Practical Applications of Medical Geology*; 2021. pp. 193–213.
42. Hein KAA. Geology of the Ranger Uranium Mine, Northern Territory, Australia: structural constraints on the timing of uranium emplacement. *Ore Geol Rev*. 2002;20(3–4):83–108.
43. Höffken J, Ramana MV. Nuclear power and environmental injustice. *Wiley Interdisciplinary Reviews: Energy Environ*. 2024; 13(1), e498.
44. Hoover E, Cook K, Plain R, Sanchez K, Waghiyi V, Miller P, Carpenter DO. Indigenous peoples of North America: environmental exposures and reproductive justice. *Environ Health Perspect*. 2012;120(12):1645–9.
45. Ian L, Aden M, Yanis M. Australia's world-leading uranium endowment and global supply and demand trends. *ASEG Ext Abstracts*. 2006;2006(1):1–4. <https://doi.org/10.1071/ASEG2006ab091>.
46. Ingram JC, Dawson TP, Whittaker RJ. Mapping tropical forest structure in southeastern Madagascar using remote sensing and artificial neural networks. *Remote Sens Environ*. 2005;94(4):491–507.
47. Jensen T, Sandström J. Fly-in/fly-out and the fragmentation of communities: A case study of a uranium mine on indigenous land. *J Rural Stud*. 2020;78:78–86.
48. Karlsson BG. Nuclear lives: Uranium mining, indigenous peoples, and development in India. *Economic and Political Weekly*; 2009. pp. 43–9.
49. Kurtio P, Auvinen A, Salonen L, Saha H, Pekkanen J, Mäkeläinen I, Komulainen H. Renal effects of uranium in drinking water. *Environ Health Perspect*. 2002;110(4):337–42.
50. Krüti P, Törnblom K, Wallimann-Helmer I, Stauffacher M. Distributive versus procedural justice in nuclear waste repository siting. *The Ethics of Nuclear Energy: Risk, Justice and Democracy in the Post Fukushima Era*. 2015; 119–140.
51. Law/Leggett RW. The behavior and chemical toxicity of U in the kidney: a reassessment. *Health Phys*. 1989;57(3):365–83.
52. Lawrence R, O'Faircheallaigh C. Ignorance as strategy: 'Shadow places' and the social impacts of the ranger uranium mine. *Environ Impact Assess Rev*. 2022;93:106723.
53. Lecomte A, Cathelineau M, Deloule E, Brouand M, Peiffert C, Loukola-Ruskeeniemi K, et al. Uraniferous bitumen nodules in the Talvivaara Ni-Zn-Cu-Co deposit (Finland): influence of metamorphism on uranium mineralization in black shales. *Mineralium Deposita*. 2014;49(4):513–33. <https://doi.org/10.1007/s00126-013-0502-3>
54. Leung CC, Yu ITS, Chen W. Silicosis Lancet. 2012;379(9830):2008–18.
55. Lottermoser B, Lottermoser BG. Mine water. Berlin Heidelberg: Springer; 2010. pp. 119–203.
56. Majumdar K, Chhandama L, Kumari P, Chatterjee D. Effects of uranium mining on health: a case study of Jadugoda of Jharkhand, India. *Discover Global Soc*. 2025a; 3(1):1–18.
57. Majumdar K, Dhaske G, Chatterjee D. Ecopolitics, Extractivism and Indigenous People: Development Paradox of the Asur Tribe in Jharkhand, India. *Int J Community Social Dev*. 2025b;7(4):439–61. <https://doi.org/10.1177/25166026251367989>.
58. Malin SA, Petzelka P. Left in the dust: uranium's legacy and victims of mill tailings exposure in Monticello, Utah. *Society and Natural Resources*. 2010;23(12):1187–1200. <https://doi.org/10.1080/08941920903005795>
59. Malin SA, Alexis-Martin B. Assessing the state of uranium research: Environmental justice, health, and extraction. *Extr Ind Soc*. 2020;7(2):512–6. <https://doi.org/10.1016/j.exis.2020.02.006>.
60. Malin SA, Ryder S, Lyra MG. Environmental justice and natural resource extraction: intersections of power, equity and access. *Environ Sociol*. 2019;5(2):109–16 <https://doi.org/10.1080/23251042.2019.1608420>.
61. Malin SA. The price of nuclear power: Uranium communities and environmental justice. Rutgers University Press; 2015.
62. McKie RE. An environmental harm perspective to examine our understanding of UK nuclear energy expansion. *Extr Ind Soc*. 2020;7(2):556–64. <https://doi.org/10.1016/j.exis.2019.10.011>.
63. Melber H. In: Young-Chan K, editor. *China and Namibia: An All-Weather Friendship Investigated*. London, UK: China and Africa; 2017. pp. 219–41.
64. Méndez-Barrientos LE, Shah SH, Roque AD, Macclements V, Stern AK. Assessing environmental justice contributions in research and public policy: an applied framework and methodology. *J Environ Policy & Planning*. 2024;26(2):188–204. <https://doi.org/10.1080/1523908X.2024.2321183>.
65. Moore JDL. Uranium Mining in Southern Africa. *South Africa and Nuclear Proliferation: South Africa's Nuclear Capabilities and Intentions in the Context of International Non-Proliferation Policies*. London: Palgrave Macmillan UK; 1987. pp. 70–81. https://doi.org/10.1007/978-1-349-07828-8_4.
66. Moore-Nall A. The legacy of uranium development on or near Indian reservations and health implications rekindling public awareness. *Geosciences*. 2015;5(1):15–29. <https://doi.org/10.3390/geosciences5010015>.
67. Nilsson JA, Randhem J. (2008). Environmental impacts and health aspects in the mining industry- A comparative study of the mining and extraction of uranium, copper and gold. Accessed from <https://publications.lib.chalmers.se/records/fulltext/85984.pdf> on August 29, 2025.

68. O'Faircheallaigh C. Indigenous peoples and mining: A global perspective. Oxford University Press; 2023. <https://doi.org/10.1093/oso/9780192894564.001.0001>.
69. Panikkar B, Brugge D. The Ethical Issues in Uranium Mining Research in the Navajo Nation. *Account Res.* 2007;14(2):121–53. <https://doi.org/10.1080/08989620701290473>.
70. Pathak M, Slade R, Pichs-Madruga R, Ürge-Vorsatz D, Shukla R, Skea J. (2022). Climate change 2022 mitigation of climate change: Technical summary.
71. Pereira R, Barbosa S, Carvalho FP. Uranium mining in Portugal: a review of the environmental legacies of the largest mines and environmental and human health impacts. *Environ Geochem Health.* 2014;36:285–301.
72. Petrova S, Marinova D. Social impacts of mining: Changes within the local social landscape. *Rural Soc.* 2013;22(2):153–65.
73. Ramadan RS, Dawood YH, Yehia MM, Gad A. Environmental and health impact of current uranium mining activities in southwestern Sinai. *Egypt Environ Earth Sci.* 2022;81(7):213. <https://doi.org/10.1007/s12665-022-10341-9>.
74. Rodon T, Thériault S, Keeling A, Bouard S, Taylor A. Mining and Indigenous Livelihoods: Rights, revenues, and resistance. Volume 344. Taylor & Francis; 2025. <https://doi.org/10.4324/9781003406433>.
75. Rodgers K, Ingram D. Decolonizing environmentalism in the Arctic? Greenpeace, complicity and negotiating the contradictions of solidarity in the Inuit Nunangat. *Am J Manage.* 2020; 20(1).
76. Reilly A. Our Territory is Not for Sale: Indigenous Led Anti-Extraction Social Movements in the Ecuadorian Amazon. *The Yale Review of International Studies*; 2020.
77. Richardson DB, Wing S, Schroeder J, Schubauer-Berigan MK, Hoffman W. Mortality among uranium miners in North America and Europe: The Pooled Uranium Miners Analysis (PUMA). *Cancer Causes Control.* 2012;23(4):553–63. <https://doi.org/10.1093/ije/dyaa195>.
78. Ruedig E, Johnson TE. An evaluation of health risk to the public as a consequence of in situ uranium mining in Wyoming, USA. *J Environ Radioact.* 2015;150:170–8. <https://doi.org/10.1016/j.jenvrad.2015.08.004>.
79. Sahoo AK, Sahu NC, Sahoo D. Impact of policy reforms on the productivity growth of Indian coal mining: a decomposition analysis. *Resour Policy.* 2018;59:460–7. <https://doi.org/10.1016/j.resourpol.2018.08.019>.
80. Sarangi AK. Uranium Resources and Production: Global Availability and Indian Scenario. *J Geol Soc India.* 2022;98(7):877–82. <https://doi.org/10.1007/s12594-022-2090-2>.
81. Sarkar A. Environmental impact assessment of uranium mining on indigenous land in Labrador (Canada): biases and manipulations. *Environ Justice.* 2019;12(2):61–8. <https://doi.org/10.1089/env.2018.0036>.
82. Schulz D, Coenen J, Lima MB, Berning L, Böörner J, Cramm M, et al. Not an easy ride: Economic research priorities for pro-environmental trade regulation. *Forest Policy and Economics.* 2026;182:103682. <https://doi.org/10.1016/j.forpol.2025.103682>.
83. Sethy N, Tripathi R, Jha V, Sahoo S, Shukla A, Puranik V. Assessment of Natural Uranium in the Ground Water around Jaduguda Uranium Mining Complex. *India Journal of Environmental Protection.* 2011;2:1002–7. <https://doi.org/10.4236/jep.2011.2.127115>.
84. Shannak SD, Cochrane L, Bobarykina D. Global uranium market dynamics: analysis and future implications. *Int J Sustain Energ.* 2025;44(1):2457376. <https://doi.org/10.1080/14786451.2025.2457376>.
85. Sheoran V, Sheoran AS, Poonia P. Soil reclamation of abandoned mine land by revegetation: a review. *Int J soil sediment water.* 2010;3(2):13.
86. Simon O, Garnier-Laplace J. Laboratory and field assessment of uranium trophic transfer efficiency in the crayfish *Orconectes limosus* fed the bivalve *C. fluminea*. *Aquatic toxicology.* 2005;74(4):372–83. <https://doi.org/10.1016/j.aquatox.2005.06.010>.
87. Singh K, Kalirajan K. A decade of economic reforms in India: the mining sector. *Resour Policy.* 2003;29(3–4):139–51. <https://doi.org/10.1016/j.resourpol.2004.08.001>.
88. Singh PK, Singh RS, Singh S. Environmental and social impacts of mining and their mitigation. In the National Seminar ESIMM-2016, Carbon sequestration through revegetated mine wasteland. 2016.
89. Sonowal CJ. Uranium Mining and Public Health Issues: Reassessing the Contention between Scientific Evidence and People's Experience in Jadugoda, Jharkhand, India. *J Social Sci.* 2020;48(4):425–37.
90. Sonowal CJ, Jojo SK. Radiation and tribal health in Jadugoda: The contention between science and sufferings. *Stud Tribes Tribals.* 2003;1(2):11–26.
91. Sovacool BK. Contesting the future of nuclear power: a critical global assessment of atomic energy. World Scientific; 2011. <https://doi.org/10.1142/7895>.
92. Srivastava RR, Pathak P, Perween M. Environmental and health impact due to uranium mining. *Uranium in Plants and the Environment.* Cham: Springer International Publishing; 2019. pp. 69–89.
93. Talukdar R. Politics and Resistance of Coal in Australia and India: Climate Justice Activism in the Global North and South. Taylor & Francis; 2025. <https://doi.org/10.4324/9781003410416>.
94. Tomasek L, Rogel A, Tirmarche M, Mitton N, Laurier D. Lung cancer in French and Czech uranium miners: radon-associated risk at low exposure rates and modifying effects of time since exposure and age at exposure. *Radiat Res.* 2008;169(2):125–37. <https://doi.org/10.1667/rr0848.1>.
95. Tornel C. Development as terricide: sacrifice zones and extractivism as state policy in Mexico. *Globalizations.* 2024; 1–25. <https://doi.org/10.1080/14747731.2024.2424075>.
96. Tsosie R. Indigenous peoples and the ethics of remediation: Redressing the legacy of radioactive contamination for native peoples and native lands. *Santa Clara J Int'l L.* 2015;13:203.
97. Tumanshiyev DA, Issova LT, Abduali MA. The geopolitical significance of kazakhstan's uranium reserves in the global security. *XABAPUJbl.* 2024;4:4.
98. Voyles TB. Wastelanding: Legacies of uranium mining in Navajo country. U of Minnesota; 2015.
99. Wang Z, Xing W. Study on the characteristics and evolution trends of global uranium resource trade from the perspective of a complex network. *Sustainability.* 2022;14(22):15295.
100. Watson JE, Joseph LN, Fuller RA. Mining and conservation: implications for Madagascar's littoral forests. *Conserv Lett.* 2010;3(4):286–7.
101. Waggitt PW. A history of Uranium mill tailings management in northern Australia. In *Proceedings, WM'04 Conference.* 2004.

102. Werner TT, Bebbington A, Gregory G. Assessing impacts of mining: Recent contributions from GIS and remote sensing. *Extractive Industries Soc.* 2019;6(3):993–1012.
103. Winde F, Brugge D, Nidecker A, Ruegg U. Uranium from Africa—An overview on past and current mining activities: Reappraising associated risks and chances in a global context. *J Afr Earth Sc.* 2017;129:759–78.
104. World Nuclear Association. World Uranium Mining Production. 2025 <https://world-nuclear.org/information-library/nuclear-fuel-cycle/mining-of-uranium/world-uranium-mining-production>. Accessed 16 Sep 2025.

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